

Charcot-Marie-Tooth Disease Type 1E (CMT1E): Comprehensive Disease Characterization Report

Target Disease: Charcot-Marie-Tooth Disease Type 1E **MONDO ID:** MONDO:0007311 | **OMIM:** #118300 | **Gene:** *PMP22* (HGNC:9118, 17p12) **Category:** Rare inherited demyelinating peripheral neuropathy

Summary

Charcot-Marie-Tooth disease type 1E (CMT1E) is a rare, autosomal dominant demyelinating hereditary motor and sensory neuropathy caused by coding **point mutations and small insertions/deletions (indels) in the *PMP22* gene** on chromosome 17p12. It is molecularly distinct from its two more common allelic disorders that arise from copy-number changes at the same locus: **CMT1A** (1.4-Mb *PMP22* duplication) and **hereditary neuropathy with liability to pressure palsies / HNPP** (*PMP22* deletion). Clinically, CMT1E is frequently **more severe than CMT1A** and is distinguished by a high frequency of **sensorineural hearing loss** (>40% of patients in a large Korean cohort), with a phenotypic spectrum that extends into the severe, early-onset Dejerine-Sottas syndrome (DSS) range. Point mutations in *PMP22* account for **less than 5% of all CMT1 cases**, making CMT1E a genuinely rare subtype within an already uncommon disease group (overall CMT prevalence ~1 in 2,500).

Mechanistically, CMT1E is a **protein-misfolding / toxic gain-of-function (and dominant-negative) disorder**. Most pathogenic substitutions fall within the conserved transmembrane domains of the tetraspan membrane protein PMP22. Mutant PMP22 misfolds, fails to traffic to the plasma membrane, and is **retained in the endoplasmic reticulum (ER) and Golgi apparatus**, where it forms **aggresome-like structures** surrounded by molecular chaperones and lysosomes. These aggregates are cleared by **autophagy**, and their accumulation drives **Schwann-cell proteostatic stress, apoptosis, and mis-trafficking of cholesterol** (retained with mutant PMP22 in the Golgi, reducing plasma-membrane cholesterol needed for myelin).

The downstream consequence is **dysmyelination/demyelination of peripheral nerves** with secondary length-dependent axonal loss — producing the classic distal, symmetric, progressive sensorimotor phenotype.

There is **no approved disease-modifying therapy** for CMT1E (or any CMT). Management is entirely supportive — rehabilitation, orthotics, corrective surgery, symptom control, and genetic counseling. Because CMT1E is a point-mutation/proteostasis disease, therapies developed for the *gene-dosage* disorder CMT1A (e.g., ascorbic acid, *PMP22* silencers) are not directly applicable; instead, **proteostasis-directed strategies** — exemplified preclinically by curcumin, which partially rescues the Trembler-J mouse — are the mechanism-matched experimental leads. The faithful natural mouse models **Trembler (Tr)** and **Trembler-J (TrJ)** carry *Pmp22* point mutations (TrJ carries the human CMT1E Leu16Pro substitution) and recapitulate the dysmyelinating pathology, providing a strong platform for mechanistic and therapeutic study.

Key Findings

Finding 1 — CMT1E is defined by *PMP22* point mutations and is more severe than CMT1A, often with deafness

CMT1E is caused by coding **point mutations or small indels in *PMP22***, in contrast to CMT1A (whole-gene duplication) and HNPP (whole-gene deletion). In a Korean cohort of 850 CMT families that were negative for the 17p12 duplication, whole-exome and targeted sequencing identified **14 pathogenic/likely pathogenic *PMP22* point mutations across 21 families**. Most mutations localized to the **conserved transmembrane domains** of the protein. Critically, **>40% of CMT1E patients showed hearing loss**, and both the physical and electrophysiological severity of CMT1E exceeded that of CMT1A, more closely resembling MPZ-related CMT1B ([PMID: 35886002](#)). As the authors state: *"point mutations or small insertions and deletions (indels) usually cause CMT type 1E (CMT1E) or HNPP" and "more than 40% of the CMT1E patients showed hearing loss. Physical and electrophysiological symptoms of the CMT1E patients were more severely damaged than those of CMT1A."*

The rarity of this mechanism is quantified: *"Point mutations in the Peripheral Myelin Protein 22 (*PMP22*) gene comprise less than 5% of the Charcot-Marie-Tooth (CMT) type 1 cases"* ([PMID: 34332267](#)).

Synonyms / alternative names: CMT1E; Charcot-Marie-Tooth neuropathy type 1E; CMT1 with deafness; PMP22-related CMT1 (point mutation type); demyelinating CMT1E. The severe end of the *PMP22* point-mutation spectrum overlaps clinically with **Dejerine-Sottas syndrome (DSS)** and congenital hypomyelinating neuropathy. Information is derived from aggregated disease-level resources (OMIM, Orphanet) combined with clinical cohort and family studies rather than EHR-derived individual data.

Finding 2 — Mechanism: mutant PMP22 misfolds, is retained in the ER/Golgi, forms aggresomes cleared by autophagy, and disrupts Schwann-cell cholesterol trafficking

The core pathophysiology is a **proteostatic / trafficking defect**. In Trembler-J (TrJ, Leu16Pro PMP22) mouse nerves, PMP22 has an **extended half-life** and forms **aggresome-like structures surrounded by molecular chaperones and lysosomes**; aggresome clearance is **enhanced by autophagy activation and blocked by autophagy inhibition** (PMID: 14627652): *"PMP22 has an extended half-life and forms aggresome-like structures that are surrounded by molecular chaperones and lysosomes" via "a mechanism that is enhanced when autophagy is activated and is primarily prevented when autophagy is inhibited."*

The trafficking lesion is specific to the transmembrane domains: *"disease-related missense mutations within transmembrane domains of PMP22 disturb intracellular protein trafficking leading to accumulation of the mutant protein in the endoplasmic reticulum/Golgi compartment"* (PMID: 10625337).

A key downstream consequence is disrupted lipid handling: in TrJ Schwann cells, **cholesterol is retained with TrJ-PMP22 in the Golgi**, with a corresponding reduction in plasma-membrane cholesterol, and cholesterol supplementation improved myelination in *PMP22*-deficient DRG explants (PMID: 32511821): *"cholesterol was retained with TrJ-PMP22 in the Golgi, alongside a corresponding reduction in its plasma membrane level."* Because cholesterol is an essential structural lipid of compact myelin, its sequestration provides a direct biochemical link between the misfolded protein and the failure of myelination.

Finding 3 — Epidemiology: CMT is the commonest inherited neuropathy (~1/2,500); CMT1E is a rare PMP22-point-mutation subtype

Overall CMT population prevalence is **~1 in 2,500** (PMID: 27584852; PMID: 38784904): *"Charcot-Marie-Tooth disease (CMT) is one of the commonest inherited neuromuscular diseases with a population prevalence of 1 in 2500."* Within CMT, demyelinating forms dominate. In the Italian

National CMT Registry (n=1,012 registered; 711 genetically diagnosed), *"Demyelinating CMT (65.3%) was more common than axonal CMT2 (24.6%) and intermediate CMT (9.0%). The PMP22 duplication was the most frequent mutation (45.2%)"* (PMID: 37170966). The same registry ranked CMT1E among the **most disabling subtypes**: *"CMT4A was the most disabling type, followed by CMT4C and CMT1E."*

CMT1E's rarity is reinforced by cohort data: point mutations comprise <5% of CMT1 (PMID: 34332267), and in some populations (e.g., a Chinese Han cohort of 77 CMT1 families) **no point mutations were detected** among duplication/deletion-negative probands (PMID: 25522693).

Disorder	PMP22 lesion	Frequency in CMT1	Relative severity
CMT1A	1.4-Mb duplication (gene dosage ↑)	~45% (most common)	Classic, typically milder
CMT1E	Point mutation / small indel	<5%	More severe; often with deafness
HNPP	Deletion (gene dosage ↓)	—	Episodic pressure palsies

Finding 4 — Trembler and Trembler-J mice are faithful natural point-mutation models; the human Trembler mutation causes Dejerine-Sottas with deafness

The Trembler-J mouse is a **natural model carrying a genuine human CMT1E mutation**: *"Trembler J (TrJ) mice carry the same Leu16Pro amino acid substitution in the PMP22 protein that is present in families diagnosed with CMT1E"* (PMID: 42105803). TrJ mice display demyelination, aggresome formation, extracellular-matrix remodeling with MMP-2/-9-expressing macrophage infiltration (PMID: 12421361), and respiratory/phrenic-nerve myelin defects with axonal atrophy (p<0.0001) plus diaphragm remodeling (PMID: 42105803).

Comparative profiling shows that point mutations act through a **distinct mechanism from gene-dosage changes**: across *Pmp22*-null, *Pmp22*-overexpressing, and Trembler mice, *"the three Pmp22 mutants exhibited distinct profiles of gene expression, suggesting different disease mechanisms"* (PMID: 15755691). This is a central caveat for therapy translation (see Finding 7).

In humans, the classic **Trembler (Gly150Asp)** substitution produces severe disease: *"We report two cases of DSS [Dejerine-Sottas syndrome] in a mother and son with the Trembler mutation, with associated findings of hearing loss and cognitive impairment"* (PMID: 35974257), underscoring that *PMP22* point mutations can drive the severe end of the demyelinating spectrum.

Finding 5 — No disease-modifying therapy exists; management is supportive; curcumin rescues the Trembler-J phenotype preclinically

"Disease-modifying therapies are not yet available for any form of CMT. Management includes rehabilitative approaches such as muscle strength training and orthotic devices, surgical interventions, symptom relief and anticipatory monitoring of associated complications" (PMID: 41571707).

The mechanism-matched preclinical lead is **curcumin**, which targets the ER-retention/aggregation-induced apoptosis specific to mutant *PMP22*: *"oral administration of curcumin partially mitigates the severe neuropathy phenotype of the Trembler-J mouse model in a dose-dependent manner. Administration of curcumin significantly decreases the percentage of apoptotic Schwann cells and results in increased number and size of myelinated axons"* (PMID: 17701891). Emerging strategies under investigation for the broader CMT field include *PMP22* silencers (for CMT1A), HDAC6 inhibitors, and unfolded-protein-response modulation (PMID: 40014417).

Finding 6 — Clinical picture and diagnosis: length-dependent progressive sensorimotor demyelinating neuropathy; NCS + genetic testing

"All forms of CMT result in length-dependent, progressive axonal loss that correlates with clinical impairments such as distal upper and lower limb weakness, musculoskeletal deformity, absent deep tendon reflexes and distal sensory deficits" (PMID: 41571707). Characteristic features include **pes cavus, hammertoes, foot drop, distal muscle atrophy, and reduced quality of life** across physical, emotional, and social domains (physical being most disabling).

Diagnosis requires *"a detailed family history, comprehensive clinical evaluation, nerve conduction studies, and relevant genetic testing"* (PMID: 42563944). Demyelinating CMT1 shows **uniformly slowed motor nerve conduction velocity (typically <38 m/s)**. CMT1E is confirmed by identifying a *PMP22* point mutation/indel **after the *PMP22* duplication has been excluded** — usually via MLPA (to exclude duplication/deletion) followed by Sanger sequencing or NGS

gene panels / WES (PMID: 35886002; PMID: 32506583). Pediatric CMT prevalence estimates range from **9.7 to 82 per 100,000** (PMID: 42563944). CMT can also involve smooth muscle, causing diaphragmatic and gastrointestinal (esophageal) dysmotility (PMID: 40270660).

Finding 7 — Genetics: autosomal dominant with frequent de novo mutations, mutational hotspots, and markedly variable expressivity

CMT1E follows **autosomal dominant** inheritance. In the Korean cohort, *"High frequencies of de novo mutations were observed, and the mutation sites of c.68C>G and c.215C>T were suggested as the mutational hotspots"* (PMID: 35886002). Most mutations lie in conserved transmembrane domains and produce a **bimodal early-onset-severe / late-onset-mild** phenotype.

Expressivity is highly variable, even within a single family: a Portuguese family carrying **p.Trp28Cys** ranged *"from asymptomatic to mild complaints of distal limb numbness and gait difficulties, with the age of onset of symptoms ranging from mid-twenties to late-sixties"* and without disability or hearing loss (PMID: 34332267) — in stark contrast to the severe DSS-with-deafness produced by the Trembler mutation (PMID: 35974257). This demonstrates that the specific variant, and its position within the protein, is a major determinant of severity.

Importantly, **dosage-model therapies do not translate to point-mutation disease**: ascorbic acid, which benefited CMT1A transgenic mice, *"in three 1-year trials... had no benefit in human beings"* (PMID: 21393063), and high-quality evidence confirms no meaningful benefit in CMT1A (PMID: 26662471). This negative result cautions strongly against extrapolating CMT1A-directed therapies to CMT1E.

Finding 8 — Anatomical/cellular targeting: Schwann-cell-autonomous demyelination of peripheral nerves, ER/Golgi as the subcellular site

The primary affected cell is the **myelinating Schwann cell** (CL:0002573); the structural target is **compact PNS myelin** (GO:0043209), with mutant PMP22 pathology localized to the **endoplasmic reticulum** (GO:0005783) and **Golgi** (GO:0005794) and cleared via **lysosomes/autophagosomes** (PMID: 14627652; PMID: 10625337; PMID: 32511821). Affected anatomy is the **peripheral nervous system** (UBERON:0000044) — including the **sciatic** (UBERON:0001322) and **phrenic** (UBERON:0002385) nerves — in a **bilateral, symmetric, length-dependent** distribution, with secondary axonal loss, musculoskeletal deformity, auditory (sensorineural hearing loss) involvement, and uncommon respiratory/GI smooth-muscle involvement (PMID:

41571707; PMID: 42105803; PMID: 40270660). PMP22 additionally **modulates store-operated calcium channel activity**, implicating altered Schwann-cell Ca^{2+} handling in the mechanism (PMID: 31213528).

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **germline (or frequently de novo) heterozygous point mutation / small indel in PMP22** (17p12), typically within a conserved transmembrane domain (hotspots c.68C>G, c.215C>T), **leads to** production of a misfolded PMP22 protein. *(demonstrated —*

P 35886002
2. Misfolded PMP22 **fails to traffic** to the Schwann-cell plasma membrane and **is retained in the ER and Golgi** apparatus. *(demonstrated —*

P 10625337
3. ER/Golgi-retained mutant PMP22 has an **extended half-life** and **aggregates into aggresome-like structures** surrounded by chaperones and lysosomes. *(demonstrated in model —*

P 14627652
4. Aggresomes are **cleared by autophagy**; when this proteostatic capacity is overwhelmed, **Schwann-cell stress accumulates**. *(demonstrated in model —*

P 14627652
5. In parallel (branch), mutant PMP22 **sequesters cholesterol in the Golgi, reducing plasma-membrane cholesterol** required for compact myelin assembly. *(demonstrated in model —*

P 32511821
6. Proteostatic stress and lipid mis-trafficking **result in Schwann-cell apoptosis** and impaired myelination. *(demonstrated in model; apoptosis reversible by curcumin —*

P 17701891

7. Additional contributing branch: altered PMP22-dependent **store-operated Ca^{2+} handling** and **MMP-2/-9-driven extracellular-matrix remodeling** (macrophage-mediated) **contribute to** nerve pathology. (*inferred / partly demonstrated —*

P 31213528

P 12421361

8. Schwann-cell dysfunction **causes dysmyelination/demyelination** of peripheral nerves (compact myelin, GO:0043209). (*demonstrated*)

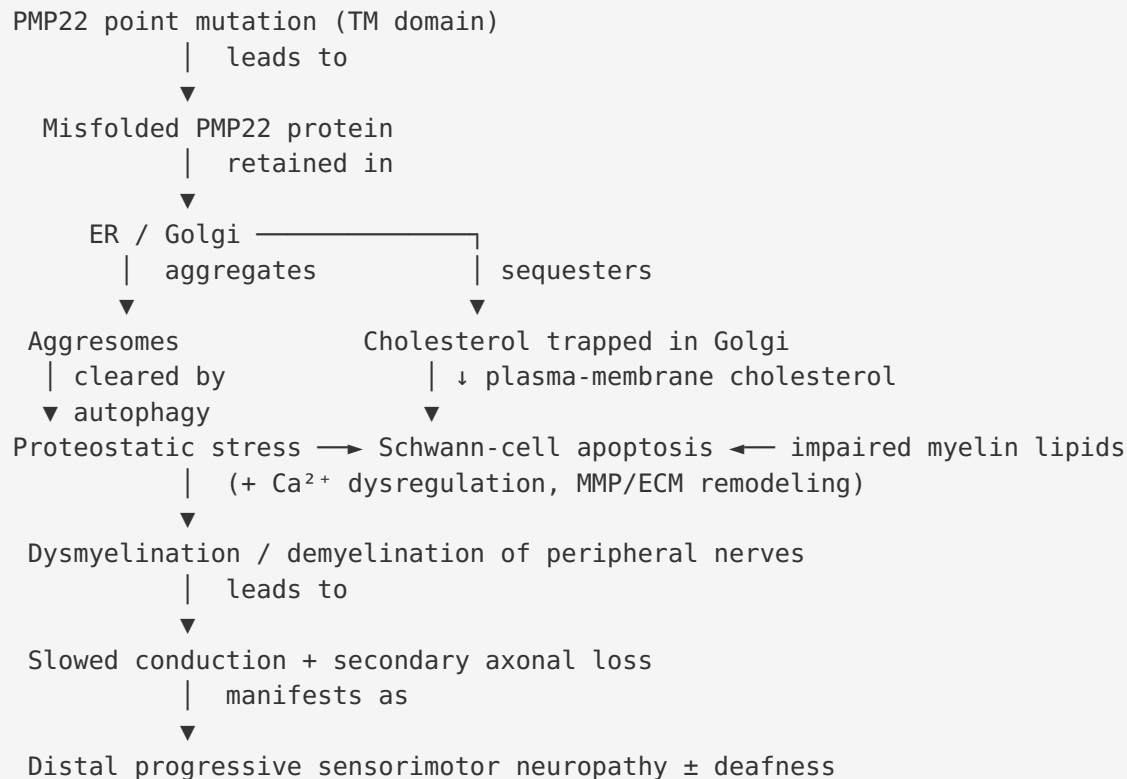
9. Demyelination **leads to** uniformly slowed nerve conduction and **secondary, length-dependent axonal degeneration**. (*demonstrated —*

P 41571707

10. Axonal loss **manifests clinically** as distal, symmetric, progressive weakness and sensory loss, pes cavus/hammertoes, areflexia, foot drop — and, in many CMT1E patients, **sensorineural hearing loss** (a distinguishing feature). (*demonstrated —*

P 35886002

P 41571707



Upstream vs downstream: The mutation and protein misfolding/ER-Golgi retention are the **upstream initiating events**; aggresome formation, cholesterol sequestration, Ca^{2+} dysregulation, and ECM remodeling are **intermediate** Schwann-cell-autonomous processes; demyelination, secondary axonal loss, and the clinical phenotype are **downstream**. The mechanism is best classified as a **toxic gain-of-function / dominant-negative** proteostasis disease — fundamentally different from the **gene-dosage** mechanism of CMT1A and HNPP, which is why CMT1A-directed therapies (ascorbic acid, *PMP22*-lowering) are not expected to help CMT1E.

Cell types (CL): myelinating Schwann cell (CL:0002573); infiltrating macrophage (CL:0000235). **Biological processes (GO):** myelination (GO:0042552); ensheathment of neurons / compact myelin (GO:0043209); ER-associated protein catabolic process / ERAD; autophagy (GO:0006914); protein folding (GO:0006457); cholesterol transport (GO:0030301); apoptotic process (GO:0006915). **Cellular components (GO):** ER (GO:0005783); Golgi apparatus (GO:0005794); lysosome (GO:0005764); aggresome (GO:0016235).

Section-by-Section Findings

1. Disease Information

CMT1E is an autosomal dominant demyelinating hereditary motor and sensory neuropathy caused by *PMP22* coding point mutations. **Identifiers:** MONDO:0007311; OMIM #118300; gene *PMP22* (HGNC:9118); chromosome 17p12. Orphanet groups it under CMT1/PMP22-related neuropathy. **Synonyms:** Charcot-Marie-Tooth neuropathy type 1E; CMT1 with deafness; PMP22 point-mutation CMT1. Information derives from aggregated disease-level resources plus clinical cohort/family studies, not EHR data.

2. Etiology

Primary cause: heterozygous germline (frequently de novo) point mutation or small indel in *PMP22*, usually in a transmembrane domain ([PMID: 35886002](#)). **Genetic risk factors:** the specific pathogenic *PMP22* variant is causal and largely deterministic; the exact variant is the main modifier of severity (variable expressivity — [PMID: 34332267](#)). **Environmental risk/protective factors and gene-environment interactions:** none established; this is a monogenic Mendelian disorder. Neurotoxic drugs (e.g., vincristine) are a general precaution in CMT but not specifically characterized for CMT1E. No protective alleles identified.

3. Phenotypes

Phenotype	Type	HPO term	Onset / severity / frequency
Distal muscle weakness/ atrophy	Clinical sign	HP:0003693 / HP:0003701	Childhood–adult; progressive; near-universal
Distal sensory loss	Symptom/sign	HP:0002936	Progressive
Pes cavus	Physical manifestation	HP:0001761	Common
Hammertoes / foot deformity	Physical manifestation	HP:0001765	Common
Areflexia / hyporeflexia	Clinical sign	HP:0001315	Common
Foot drop / steppage gait	Clinical sign	HP:0009830	Common
Sensorineural hearing loss	Sign / lab (audiometry)	HP:0000407	>40% of CMT1E (PMID: 35886002) — distinguishing
Slowed motor NCV (<38 m/s)	Lab / electrophysiology	HP:0003431	Universal in demyelinating CMT1
Respiratory / diaphragmatic involvement	Sign	HP:0002093	Uncommon (PMID: 42105803 , model)
Esophageal / GI dysmotility	Symptom	HP:0002015	Rare (PMID: 40270660)
Cognitive impairment (severe/DSS end)	Sign	HP:0001268	Rare, severe variants (PMID: 35974257)

Onset: ranges from congenital/infantile (severe, DSS-like) to late-adult (mild), reflecting bimodal expressivity. **Progression:** slowly progressive, chronic. **QoL:** reduced across physical, emotional, and social domains, with the physical domain most disabling ([PMID: 41571707](#)).

4. Genetic / Molecular Information

Causal gene: *PMP22* (OMIM 601097; HGNC:9118), encoding a 22-kDa tetraspan integral membrane protein of compact PNS myelin ([PMID: 11002284](#)). Variant types: missense (predominant, transmembrane domains), plus small indels; classified pathogenic/likely pathogenic per ACMG/AMP. Representative variants: hotspots c.68C>G and c.215C>T ([PMID: 35886002](#)); p.Trp28Cys (mild, Portuguese family — [PMID: 34332267](#)); Leu16Pro (TrJ / CMT1E); Gly150Asp (Trembler / DSS — [PMID: 35974257](#)). Allele frequency: pathogenic variants are private/rare (absent or extremely rare in gnomAD). Origin: germline; frequent de novo. Functional consequence: toxic gain-of-function / dominant-negative via misfolding, not haploinsufficiency. Modifier genes/epigenetics: none established; the causal variant itself is the principal severity determinant. Chromosomal abnormalities:* none (point-mutation disease; distinct from the CNVs underlying CMT1A/HNPP).

5. Environmental Information

No environmental, lifestyle, or infectious contributors are established. CMT1E is monogenic and Mendelian. General neurotoxin avoidance (e.g., vincristine) applies to CMT broadly.

6. Mechanism / Pathophysiology

See the **Mechanistic Model** section above for the ordered causal chain, branch points, and ontology terms. Key steps: transmembrane-domain misfolding → ER/Golgi retention → aggresome formation cleared by autophagy → cholesterol sequestration and Ca²⁺ dysregulation → Schwann-cell apoptosis → dysmyelination → secondary axonal loss → clinical neuropathy with deafness.

7. Anatomical Structures Affected

Primary: peripheral nerves (UBERON:0000044), especially motor/sensory nerves of distal limbs; sciatic (UBERON:0001322) and phrenic (UBERON:0002385) nerves in models. **Body systems:** peripheral nervous system; secondary musculoskeletal (foot deformity), auditory system (sensorineural hearing loss), and occasionally respiratory/GI smooth muscle. **Cell:** myelinating Schwann cell (CL:0002573). **Subcellular:** ER (GO:0005783), Golgi (GO:0005794), lysosome, aggresome. **Laterality:** bilateral, symmetric, length-dependent.

8. Temporal Development

Onset: bimodal — early-onset severe (infantile/childhood, DSS-like) vs late-onset mild (adult, up to late-sixties). **Course:** chronic, slowly progressive, lifelong; not episodic or relapsing-remitting (contrast HNPP). **Critical periods:** developmental myelination is when *PMP22* dysfunction is most consequential; early Schwann-cell proteostasis may be a therapeutic window (PMID: 14627652, PMID: 17701891). No spontaneous remission.

9. Inheritance and Population

Inheritance: autosomal dominant; frequent de novo mutations (PMID: 35886002). **Penetrance:** high but variable expressivity even intrafamilially (PMID: 34332267). **Prevalence:** CMT overall ~1/2,500 (PMID: 27584852); CMT1E <5% of CMT1 (PMID: 34332267); pediatric CMT 9.7–82/100,000 (PMID: 42563944). **Anticipation / mosaicism / founder effects:** not established for CMT1E specifically; de novo occurrence is common. **Sex ratio:** no strong sex bias reported. **Consanguinity:** not relevant (dominant).

10. Diagnostics

Electrophysiology: nerve conduction studies showing uniform motor NCV slowing (<38 m/s) — hallmark of demyelinating CMT1. **Genetic testing algorithm:** exclude *PMP22* duplication/deletion first (MLPA/CMA), then Sanger or NGS panel / WES to detect the point mutation (PMID: 32506583, PMID: 35886002). **Audiometry** for hearing loss. **Nerve biopsy** (rarely needed) shows demyelination and onion-bulb formation. **Differential diagnosis:** CMT1A (duplication), CMT1B (*MPZ*), CMTX1 (*GJB1*), CIDP (acquired), Dejerine-Sottas syndrome. **Screening:** cascade genetic testing of at-risk relatives; prenatal/preimplantation testing available given a known familial variant.

11. Outcome / Prognosis

Survival: normal or near-normal life expectancy in typical CMT1E; not directly life-limiting, though severe early-onset (DSS) forms carry higher morbidity, and rare respiratory involvement can affect prognosis (PMID: 42105803). **Morbidity:** significant, chronic disability from progressive weakness, deformity, and sensory loss; CMT1E ranks among the most disabling CMT subtypes (PMID: 37170966). **Prognostic factors:** specific variant / domain location, age of onset, presence of hearing loss. No validated molecular prognostic biomarkers.

12. Treatment

No disease-modifying therapy (PMID: 41571707, PMID: 40014417). **Supportive/rehabilitative (NCIT-type interventions):** physical therapy / strength training (NCIT: Physical Therapy), orthoses/AFOs (NCIT: Orthotic Device), occupational therapy, corrective foot surgery, pain management, hearing aids. Strength training showed benefit for timed walking in CMT (PMID: 19588347). **Not indicated:** ascorbic acid (no benefit in CMT1A; dosage-model therapy — PMID: 21393063, PMID: 26662471). **Experimental / mechanism-matched:** curcumin (proteostasis/anti-apoptotic; preclinical rescue of TrJ — PMID: 17701891); autophagy enhancers; HDAC6 inhibitors; UPR modulators (PMID: 40014417). *PMP22* silencers target CMT1A dosage and are **not** appropriate for point-mutation CMT1E unless allele-specific.

13. Prevention

No primary prevention (monogenic). **Genetic counseling** is central: autosomal dominant inheritance with 50% transmission risk, plus de novo recurrence counseling. **Reproductive options:** prenatal diagnosis and preimplantation genetic testing when the familial variant is known. **Tertiary prevention:** rehabilitation, orthotics, foot care, fall prevention, and anticipatory monitoring for respiratory/auditory complications.

14. Other Species / Natural Disease

Pmp22 is highly conserved; **mouse (NCBI Taxon:10090)** orthologue *Pmp22* underlies the spontaneous **Trembler (Tr)** and **Trembler-J (TrJ)** mutants — naturally occurring dysmyelinating neuropathies that are genuine models of human *PMP22* point-mutation disease (PMID: 42105803, PMID: 30685714). No zoonotic potential. Evolutionary conservation of *PMP22* function in myelination makes rodent models informative but imperfect (species differences in lesions — PMID: 30685714).

15. Model Organisms

Natural point-mutation mice: Trembler-J (Leu16Pro, the human CMT1E allele) and Trembler (Gly150Asp) (PMID: 42105803, PMID: 15755691). **Engineered models:** *Pmp22*-null and *Pmp22*-overexpressing lines model HNPP and CMT1A respectively, and comparative studies show **distinct expression profiles / mechanisms** by mutation type (PMID: 15755691). **Recapitulation:** TrJ faithfully reproduces dysmyelination, aggresome formation, apoptosis, ECM remodeling, and phrenic-nerve/diaphragm pathology. **Limitations:** rodent lesions differ

in some respects from human nerve pathology ([PMID: 30685714](#)). **In vitro:** DRG/Schwann-cell co-cultures and patient-derived iPSC-Schwann cells are relevant for trafficking/proteostasis assays. **Resources:** MGI, IMSR.

Evidence Base

PMID	Title (abbreviated)	Evidence type	Role
35886002	<i>PMP22 mutations in CMT1E patients</i>	Human cohort (Korea)	Defines CMT1E genetics, deafness >40%, severity vs CMT1A, de novo hotspots
34332267	<i>Portuguese family, novel PMP22 point mutation</i>	Human family	Rarity (<5% of CMT1); variable expressivity (p.Trp28Cys)
14627652	<i>Autophagy removes aggresomes in Schwann cells</i>	Model (TrJ)	Aggresome formation; autophagic clearance
10625337	<i>PMP22 and demyelinating neuropathies</i>	In vitro/ review	ER/Golgi retention of TM-domain mutants
32511821	<i>Subcellular diversion of cholesterol by PMP22</i>	Model/in vitro	Cholesterol trapped in Golgi; myelination rescue
42105803	<i>Diaphragm/phrenic defects in CMT1E mouse</i>	Model (TrJ)	TrJ = human Leu16Pro; respiratory involvement
15755691	<i>Distinct mechanisms by PMP22 dosage vs point mutation</i>	Model	Point mutation ≠ dosage mechanism
35974257	<i>DSS from Trembler mutation</i>	Human	Severe end: DSS + deafness + cognitive
17701891	<i>Oral curcumin mitigates TrJ phenotype</i>	Model	Mechanism-matched preclinical therapy
41571707	<i>CMT and related neuropathies</i>	Review	No DMT; supportive care; clinical phenotype
40014417	<i>CMT management 2025</i>	Review	Emerging therapies (silencers, HDAC6i, UPR)
42563944	<i>CMT in children</i>	Review	Diagnostic workup; pediatric prevalence
37170966	<i>Italian National CMT Registry</i>	Registry	Demyelinating 65.3%; CMT1E among most disabling

PMID	Title (abbreviated)	Evidence type	Role
27584852	<i>Recent advances in genetic neuropathies</i>	Review	CMT prevalence 1/2,500
32506583	<i>Diagnostic yield of sequencing strategies</i>	Human	MLPA→Sanger/NGS diagnostic algorithm
25522693	<i>PMP22 neuropathies in Chinese Han</i>	Human cohort	No point mutations detected — rarity
21393063 / 26662471	<i>Ascorbic acid trials</i>	RCT	Negative: dosage-model therapy fails
31213528	<i>PMP22 and store-operated Ca^{2+}</i>	In vitro	Ca^{2+} -handling contribution
12421361	<i>MMP-mediated ECM degradation in Trj</i>	Model	Macrophage/MMP ECM remodeling
11002284	<i>Membrane topology of PMP22</i>	In vitro	Tetraspan protein structure
30685714	<i>Rodent PMP22 models</i>	Review	Model relevance and limitations

Consistency of the evidence: Findings converge across independent human cohorts (Korea, Portugal, Italy, China, Brazil), natural mouse models (Trembler/Trj), and in vitro trafficking studies. The proteostasis/misfolding mechanism is supported by ≥ 4 independent model/in vitro lines. The one important *negative* result — ascorbic acid failing in CMT1A RCTs — reinforces (rather than challenges) the conclusion that CMT1E, a distinct proteostasis disorder, requires distinct, mechanism-matched therapy.

Limitations and Knowledge Gaps

- 1. CMT1E-specific epidemiology is imprecise.** No dedicated prevalence/incidence estimate exists for CMT1E; it is inferred as $<5\%$ of CMT1. Sex ratio, penetrance quantification, and natural-history data are largely extrapolated from broader CMT/CMT1 datasets.

2. **Genotype–phenotype correlation is incomplete.** Although variant position (transmembrane vs extracellular) clearly influences severity, no comprehensive map links specific *PMP22* residues to hearing loss, onset, or progression rate.
 3. **Human mechanistic data are limited.** Much of the misfolding/aggresome/cholesterol mechanism derives from Trembler/TrJ mice and in vitro systems; direct confirmation in human CMT1E nerve or iPSC-Schwann cells is sparse.
 4. **No CMT1E-specific therapeutic trials.** Curcumin data are preclinical (mouse) only; no human trials in CMT1E. Emerging therapies (HDAC6 inhibitors, UPR modulators, allele-specific silencing) remain unvalidated for point-mutation disease.
 5. **Hearing-loss mechanism unexplained.** The >40% frequency of sensorineural hearing loss is well documented but its cellular basis (cochlear vs auditory-nerve myelination) is not established.
 6. **Ca²⁺ and ECM contributions** (store-operated calcium, MMP-mediated remodeling) are documented in models but their causal weight in human disease is uncertain.
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Proposed Follow-up Experiments / Actions

1. **Genotype–phenotype registry analysis:** Aggregate ClinVar/HGMD *PMP22* point-mutation variants with clinical severity and hearing-loss status to build a residue-level severity map, distinguishing transmembrane vs loop-domain effects.
2. **Patient-derived iPSC-Schwann-cell models:** Generate CMT1E lines (e.g., Leu16Pro, Trp28Cys, Gly150Asp) to confirm ER/Golgi retention, aggresome burden, cholesterol mis-trafficking, and apoptosis in a human context, and to screen proteostasis modulators.
3. **Mechanism-matched therapeutic screening:** Test autophagy enhancers, chemical chaperones, HDAC6 inhibitors, and UPR modulators against mutant-*PMP22* clearance and myelination endpoints in TrJ mice and human iPSC-Schwann cells; advance the most efficacious to CMT1E-specific trial design.
4. **Cholesterol-supplementation study:** Given Golgi cholesterol sequestration, evaluate dietary/pharmacologic cholesterol or statin-pathway modulation on myelination in TrJ models.
5. **Allele-specific silencing:** Develop and test ASO/siRNA approaches that selectively suppress the mutant *PMP22* allele (preserving wild-type dosage) — the rational silencing strategy for a dominant point-mutation disease.

6. **Auditory phenotyping:** Systematically characterize the cochlear/auditory-nerve basis of CMT1E hearing loss (audiometry + ABR in patients; cochlear histology in models) to define an underexplored, distinguishing feature.
 7. **Natural-history cohort:** Establish a CMT1E-specific longitudinal cohort (CMTNS, timed walk, QoL, audiometry, respiratory function) to define progression rate and validate outcome measures for future trials.
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Report compiled from an autonomous multi-iteration literature investigation (32 papers reviewed; 8 confirmed findings). Evidence types span human clinical/cohort studies, natural and engineered mouse models, and in vitro systems, as annotated throughout.

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